



New York University
Computer Science Department
Introduction to Software Development
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Deadline: See NYUclasses for the deadline, 5% off per day after the deadline (3 d maximum).

Please read the guidelines carefully to avoid receiving a zero on the HW:

- Create a folder for each exercise and include all your java classes in the same folder. Name this folder ExerciseN where N is the exercise number
- Apply this convention to all future homework assignments during the semester.
- Compile and run the program for each exercises.
- Make an archive (zip file or compressed file) with all the java files (the .java files NO the .class files) and post it on NYU Classes (StudentName_Homework_X.zip)
- It is your responsibility to make sure that the Zip files have your actual files. You may send the file to yourself by email to double check before you upload on NYU classes.
- If the graders cannot open the file, you will receive a grade of zero.
- If you send the .class files instead of the .java files (source files) you will receive a zero.
- Cheating will be severely addressed with an immediate zero on the homework and a report to the academic advisor and the administration.
- You will automatically lose 50% of the points for an exercise if the program does not compile and run correctly

Exercise. 1 (50 pts)

To work in this exercise, you need to have a basic knowledge of Hexadecimal numbering system and how to convert from hex to decimals.

The following link should provide you just enough background, but feel free to do your own research if necessary. <https://www.mathsisfun.com/hexadecimals.html>

Create a java class and name it `Excercise1` which has a main method and a method called **hex2Dec**. The `hex2Dec` takes the string "hexString" as an input and returns its decimal conversion as an int. Also, the method throws a custom exception called **HexFormatException** if any character of the input is not a proper hex string.

Create the custom exception class **HexFormatException** which extends `Exception` class.

Assignment 8

Main method will call hex2Dec method and you will need to catch the exception from the main method and just print the message *"Illegal hex character"* in the console.

Your code can use the just have the following five hexadecimal numbers
A5, FAA, T10, ABC, 10A as shown in the following pseudo code illustration

```
Class Exercise1{
    main method will try/catch the exceptions
        System.out.println(hex2Dec("A5"));
        System.out.println(hex2Dec("FAA"));
        System.out.println(hex2Dec("T10"));
        System.out.println(hex2Dec("ABC"));
        System.out.println(hex2Dec("10A"));
```

hex2Dec will take single parameter (String hexString), return an int, and will throw the custom exception

<your logic to convert hex to dec and >

<you can add any supporting methods you like as long as your class contains the above two methods>

Exercise. 2 (50 pts)

Create a class and call it VotingValidator.java that does the following:

- Ask the user to enter their DoB in the format MM/DD/YYYY or ask the user to enter Day of birth, month of birth and year of birth separately.
- The class has a method named ValidateAge which accepts the user input and throws a custom exception named *InvalidAgeException*.
- ValidateAge method doesn't return anything, and it checks if the user is 21 years old by Election day (November 3rd, 2022) or not.
 - o If the user is 21 or older by 2022, the class prints a message "Welcome to Vote!"
 - o Otherwise, your method will throw *InvalidAgeException* which is the custom exception you will create. This exception will print the following message in the console when thrown "Error: User can not vote this election because user's DoB is <USER_DOB> and he/she will be <age> by election day. "
- ValidateAge method is called from the main method and is wrapped by a try/catch to handle the exception.
- PS. You will need to use the date & calendar class for age validation.